Optimizer Coupling Masks Loss-Asymmetric Shifts in Finite-Budget Coordination-Collapse Crossings*

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Paper under double-blind review

Abstract

Cooperative multi-agent reinforcement learning can collapse into low-cooperation behavior when local incentives outweigh the team objective. We define an operational estimand for where this happens: the finite-budget crossing w_{50} along a local/team reward-weighting dial at which collapse becomes more likely than not, estimated from behavioral labels with training-seed-block bootstrap intervals. Asking whether a loss-asymmetric reward transform moves it, we find the answer depends on optimizer coupling. In default Stable-Baselines3 PPO on an inventory network, actor and critic share one gradient clip that the critic dominates; scaling both reward branches by seven moves the crossing by +0.357 [+0.249, +0.464] and shrinks the actor's parameter updates to 0.39 of baseline. Clipping actor and critic separately restores the updates and removes part of the shift (−0.207 [−0.336, −0.085]); reward normalization removes the detectable remainder, consistent with a critic-target-scale effect we have not isolated. With separate optimizers and normalized rewards, where no symmetric shift is detectable (−0.049 [−0.117, +0.017]), a loss-only multiplier still moves the crossing relative to the symmetric control (+0.237 [+0.136, +0.346]), through 3.6M transitions (+0.154 [+0.053, +0.256]). In Level-Based Foraging an asymmetric transform also moves the crossing, there by penalizing a cooperative action. Crossings stay budget- and configuration-dependent: w_{50} is a comparative instrument, not a critical point.

1 Introduction

A team of learning agents can fail even when the task has a good cooperative solution: instead of learning complementary behavior, agents can settle into locally safe, low-cooperation behavior. We call this *coordination collapse*; in the MARL literature its endpoint is the *lazy-agent* policy (Liu et al., 2023; Shi & Peng, 2023), and the underlying private-versus-collective conflict is the one studied in sequential social dilemmas (Leibo et al., 2017). Credit-assignment and centralized-training methods mitigate it (Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018), and decentralized reward shaping is known not to match centralized training (Akella, 2025). Less is known about *where* collapse sets in as the reward design changes.

We study this through a local/team *reward-weighting* dial: agent i is trained on $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0,1]$, so at w=0 the learner sees only its own reward and at w=1 only the team reward. The dial is a natural experimental axis because it is the design choice practitioners actually make, and because collapse is common at one end and rare at the other. We summarize it by an operational, finite-budget 50% behavioral crossing w_{50}, estimated from pre-specified evaluation labels with seed-paired uncertainty. Weng & Lee (2026) locate an empirical collapse threshold over a payoff-harshness parameter; w_{50} plays the same role on the reward-weighting axis and lets two reward designs be compared by a paired contrast. Because the team reward sums agents' rewards, the dial changes reward scale as well as reward composition (Section 3).

Reward-transform comparisons are where scale matters. We use prospect theory only as a controlled reward-transform family, not a psychological claim: a pointwise transform of the blended reward with value-function

*Large language models were used as general-purpose assistive tools for editing, code review, and analysis scripting. The authors designed the study, verified the reported results, and take responsibility for the complete submission.

curvature and a loss-branch multiplier λ (Kahneman & Tversky, 1979; Tversky & Kahneman, 1992). Any such transform also changes the reward's magnitude, and in actor–critic learners reward magnitude reaches the policy through the optimizer. In the default Stable-Baselines3 PPO configuration actor and critic share one global gradient clip, so a larger reward can change the actor's update without changing what the reward says. We ask three questions. **RQ1:** Can a finite-budget crossing along the dial be estimated with useful precision, and how does it depend on training budget? **RQ2:** Does a symmetric rescaling of the reward move the crossing, and if so through which optimizer pathway? **RQ3:** With that pathway removed, does a loss-only multiplier still move the crossing relative to a symmetric control?

The contributions are: (i) *a measurement protocol*: observed collapse proportions, Firth logistic midpoints, bracketing checks and training-seed-block bootstrap intervals that make reward designs comparable by paired contrasts. The midpoint itself is standard response-curve machinery; the protocol's value is its controlled use and its explicit dependence on budget, reward units, optimizer and label. (ii) *an optimizer confound and its mechanism*: in pre-registered experiments (2,320 trajectories), symmetric $\times 7$ scaling moves the crossing at least as far as a loss-only multiplier under the default configuration. Parameter-update logging shows that the critic takes the actor's share of the shared clip, and per-group clipping restores the actor's updates and removes part of the shift. The remainder, which clipping does not touch, disappears under reward normalization; it is consistent with critic target scale but not isolated. (iii) *a controlled asymmetric effect*: with separate optimizers and normalized rewards no symmetric shift is detectable, while the loss-only multiplier still raises the crossing relative to the symmetric control, through 3.6M transitions and, by a different incentive route, in Level-Based Foraging. Along the way we find that crossings remain budget-dependent and non-monotone to 3.6M transitions and that neither collapse nor coordination is absorbing. We use "boundary" only in this operational sense, not as a claim of phase-like criticality.

Section 2 positions the work; Section 3 defines the reward, transforms, learner and estimand; Sections 4–6 establish the crossing, its budget dependence and the legacy transform comparisons; Section 7 isolates the optimizer pathway; Section 8 turns to a second environment; Section 9 states what the results do and do not establish.

2 Related Work

Cooperative MARL, studied since independent Q-learning on shared tasks (Tan, 1993; Buşoniu et al., 2008), inherits the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al., 2023; Shi & Peng, 2023; Tampuu et al., 2017). The standard remedy is centralized credit assignment under CTDE, a centralized critic or factorized value function (Lowe et al., 2017; Yu et al., 2022; Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018; Son et al., 2019); related work ties the failure to decentralized exploration and parameter sharing, and locates an empirical collapse threshold over payoff harshness for each exploration level, averaged over 30 seeds (Weng & Lee, 2026), or shows decentralized shaping cannot match centralized baselines (Akella, 2025). This establishes *that* centralization helps, *how* to implement it, and that collapse thresholds can be measured. It does not provide a 50% collapse-probability crossing along a local/team reward-weighting dial with seed-paired intervals, or test how reward transforms and optimizer implementation move it. Combined individual-and-team reward design has direct precedent: DE-MADDPG separates a per-agent and a team critic for such mixtures (Sheikh & Bölöni, 2020), packet-routing work studied the lazy-versus-selfish trade-off with mixed and adaptive local/global signals (Boyan & Littman, 1994; Mao et al., 2020), IRAT learns individual- and team-reward policies with mutual discrepancy constraints (Wang et al., 2022), and CLOT balances individual and team optimization through a Lagrangian multiplier (Zhang et al., 2026). These set or adapt the mixture as a design choice rather than *locating* a crossing with uncertainty.

Statistically, a 50% crossing is an effective-dose or psychometric-threshold estimand. Dose–response analysis routinely estimates ED50 values and differences between them (Ritz et al., 2015), and psychometric work studies bootstrap uncertainty and grid placement (Wichmann & Hill, 2001). We import that logic into trained multi-agent populations; our contribution lies in the experimental unit, collapse label, seed-block resampling, bracketing diagnostics and paired comparison of reward designs. Treating collective learning as a dynamical system with critical points has precedent in phase diagrams over environmental axes (Yamaguchi,

2025), metastable opinion-dynamics boundaries (Capannoli et al., 2026), and mean-field limits of multi-agent learning (Hu et al., 2019; Barfuss et al., 2019; Leonte et al., 2025); ours is a reward-structure threshold estimated from trained populations rather than asserted to be a dynamical critical point.

PPO's behavior depends on implementation details that are not part of the algorithm's definition (Engstrom et al., 2020; Andrychowicz et al., 2021), among them reward or value normalization (van Hasselt et al., 2016) and gradient clipping; cooperative PPO implementations normalize value targets by default (Yu et al., 2022). Our factorial shows that one such detail, a global clip shared by actor and critic without value normalization, changes the qualitative answer to a reward-design question.

Human preference structure has entered collective learning only narrowly. Hughes et al. (2018) show inequity aversion improves cooperation within the cooperative-AI agenda (Dafoe et al., 2020; Hammond et al., 2025; Barfuss et al., 2020); KTO aligns language models with a Kahneman–Tversky value function carrying an explicit loss-aversion coefficient (Ethayarajh et al., 2024); risk-sensitive RL optimizes tail outcomes (Qiu et al., 2021; Shen et al., 2023; Prashanth, 2014; Tamar et al., 2015); and prospect theory has entered single-agent RL with known calibration concerns (Prashanth et al., 2016; Rabin, 2000; Tversky & Kahneman, 1992). Mengesha et al. (2025) find loss aversion non-monotone in an evolutionary quantal-response model, and Lalmohammed et al. (2025) bring CPT into cooperative MARL inside the actor–critic update, varying value and probability-weighting components jointly with the optimizer. Unlike inequity aversion, which penalizes the *difference* between own and peer payoffs, our transform acts on the already-blended scalar against a fixed reference, so it changes the magnitude and sign-weighting of the training signal rather than the local-versus-collective trade-off itself.

3 Preliminaries and Method

We study *reward-mixed* cooperative MARL, in which each learner's scalar reward is the blend $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0,1]$. On the supply chain the team reward sums the five trained nodes' profits, so $x_i(w) = r_i^{\text{local}} + w \sum_{j \neq i} r_j^{\text{local}}$: node i's own profit keeps unit weight and w adds teammates' profits. We do not rescale r_i^{local} or r^{global} at each w, so a given w means the same mixture in every condition; the scale of x_i therefore varies with w, which matters under the optimizer coupling studied in Section 7. Each experiment varies one factor at a time.

The boundary estimate follows four steps: fix the environment, learner, horizon, and collapse label; train matched policies across a grid of w; label each run from pre-specified *evaluation* behavior, not training reward; and fit a monotone binary model $P(\text{collapse} \mid w)$, reporting the midpoint w_{50} with bootstrap confidence intervals.

Reward transforms. With zero as the gain/loss reference, $v(x) = x^\alpha$ for $x \geq 0$ and $-\lambda(-x)^\beta$ for $x < 0$. The exponents $\alpha, \beta \in (0,1]$ control *curvature*; the multiplier $\lambda > 1$ scales the loss branch. For $a > 0$, $|v(-a)|/v(a) = \lambda a^{\beta-\alpha}$, which equals λ at every magnitude only when $\alpha = \beta$, as in all our experiments. The transform is applied after blending,

$$r_i^{\text{PT}} = v\big((1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}\big)\,. \tag{1}$$

Controls use $r_i^{\text{PT}} = v((x_i - z_0)/s_{\text{in}})/s_{\text{out}}$ with no addition of z_0 after transformation; Equation 1 has $z_0 = 0$ and $s_{\text{in}} = s_{\text{out}} = 1$. The scale controls of Section 7 use the frozen empirical-mean reference $z_0 = -2.6281398333$ (Appendix A) with $u = x - z_0$ and $\phi(u) = \text{sign}(u)|u|^{.88}$: arm B= $\phi(u)$ (curvature only), A= $v_7(u)$ (losses $\times 7$), S= $7\phi(u)$ (both branches $\times 7$, i.e. $s_{\text{out}} = 1/7$), and R= $3.451\,\phi(u)$ (both branches scaled by A's pooled RMS ratio). We omit the probability-weighting channel of full CPT. Unlike Lalmohammed et al. (2025), the transform sits outside the optimizer, so within a fixed optimizer configuration matched runs differ only in the scalar training signal.

Learner and optimizer. The supply-chain learner is parameter-shared PPO in Stable-Baselines3 2.8.0 (Raffin et al., 2021): one policy across nodes, separate actor and critic MLPs, a single Adam optimizer ($\epsilon = 10^{-5}$, SB3's implicit default) over both, advantages normalized per minibatch, no value normalization, and one global gradient-norm clip of 0.5 applied to the concatenated actor and critic gradient. Because the

Table 1: Experimental setup. Legacy sweeps (earlier platform, 20 seeds per cell) supply the 300k decomposition and the cross-sectional horizon fit; pre-registered experiments E1–E6 run on one platform with ten seeds per cell.

	Supply chain (legacy, E1–E3)	Level-Based Foraging (E4)
Environment	Six-node `NetworkManagement-v1`; node 1 scripted, nodes 2–6 trained; 30-period episodes.	`Foraging-5x5-2p-2f-v3`; two players, two foods; 50-step episodes.
Local reward	Node i's period profit (revenue minus procurement, operating, holding, pipeline and unfulfilled-demand costs).	Native food share minus 0.02 per LOAD attempt.
Team reward	Sum of the five trained nodes' profits.	Sum of native food rewards.
Learner	Parameter-shared PPO (SB3 2.8.0); separate actor and critic MLPs.	MAPPO: actor on own observation, critic on global state (SB3 2.8.0).
Optimizer	Default: one Adam step ($\epsilon = 10^{-5}$), global clip 0.5, no value normalization. E1 adds separate no-clip optimizers and `VecNormalize`; E2 sets $\epsilon = 10^{-8}$; E6 clips actor and critic separately; E5 uses the fully decoupled learner.	Global clip 0.5 or separate no-clip optimizers.
Collapse label	$\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$ on ten deterministic evaluation episodes.	Native team return < 0.626 over ten evaluation episodes (no effort cost).
Arms	Linear; curvature $\alpha{=}\beta{=}.88$ with $\lambda \in \{1, \ldots, 7\}$ (legacy); B, A ($\lambda = 7$), S ($\times 7$), R ($\times 3.451$) with $z_0 = -2.628$ (E1, E2); linear, curvature, $\lambda = 7$ (E3); B, A, S (E5, E6).	Linear; curvature; A ($\lambda = 4$); S (both branches $\times 4$).
Budget, grid	300k, eight cells (E1, E2); 3.6M with 24 checkpoints, nine cells (E3, E5).	2M timesteps; ten cells $\{0, .1, .2, .3, .5, .65, .75, .85, .9, .95\}$.
Uncertainty	Firth logistic midpoint; 2,000 multinomial draws of complete training-seed blocks, shared across conditions so that contrasts are paired.	

actor's advantages are normalized but the critic's targets are not, a larger reward scale enlarges only the value-loss gradient, which then dominates the shared clip. Section 7 varies exactly this coupling. Training budgets count requested agent transitions; with five trained lanes and 128-step rollouts each rollout adds 640 transitions, so 300k requested transitions execute 300,160 (Appendix A).

Collapse label. A *collapse label* is a pre-specified binary evaluation rule scored on held-out evaluation episodes. In the supply chain, collapse is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, where $\bar{\pi}$ is the total profit of nodes 2–6 averaged over ten deterministic evaluation episodes (seeds $10s, \ldots, 10s + 9$ for training seed s) and $\bar{S}$ is the mean over the five trained nodes of the deterministic base-stock target $250(a_1 + 1)$ chosen at the reset of the first evaluation episode, where a_1 is the node's first action component. Node 1 is scripted to order 20 units, split equally across its action entries. Collapsed runs sit at the lazy-policy profit floor (≈ -128), so the profit clause is a redundancy check on $\bar{S} < 10$; threshold variants of $\pm 20\%$ change w_{50} by at most 0.015. Across all 1,280 E1 runs only 1 has $\bar{S}$ within one unit of the threshold, so the label is not a fine cut through a continuum (Figure 9, Appendix E).

Estimation. We write the estimand as $w_{50}(T; E, A, V, U, L, Q)$: training budget, environment, learner (including its optimizer configuration), transform, reward units, collapse label and evaluation protocol. It is a fitted 50% crossing, not a safe operating point, and a rightward shift means more team weight is needed to reach it. We fit $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$ by Firth logistic regression (Firth, 1993) and report $w_{50} = -b_0/b_1$. Intervals use 2,000 multinomial draws of complete training-seed blocks; the same draws serve every condition in an experiment, so contrasts are seed-paired (Efron & Tibshirani, 1993). Out-of-grid and non-converged draws are retained and counted, never clipped. Logistic, probit and decreasing isotonic fits on the same rows serve as sensitivity estimators (convention in Appendix B); their maximum spread on the supply-chain data is 0.018. Leaving out any one of the ten training seeds changes the sign of none of the principal contrasts (Table 12). The primary substrate is `NetworkManagement-v1` (Hubbs et al., 2020), a six-node multi-echelon inventory network (Oroojlooyjadid et al., 2022; Gijsbrechts et al., 2022; Leluc et al., 2023) wrapped with PettingZoo/SuperSuit (Terry et al., 2021; 2020); the second environment is Level-Based Foraging (Section 8).

Pre-registration and platform. Experiments E1–E6 were pre-declared, with source and design hashes frozen before launch; the departures (one pre-execution source fix, a label-threshold fallback corrected from stored data, and analysis-code fixes) are logged in Appendix C. All E1–E6 trajectories run on one CPU

platform. An earlier platform does not reproduce trained policies bit-for-bit, but collapse labels agree in 221/240 (92.1%) of matched direct-control runs and 3,289/3,600 (91.4%) of matched horizon checkpoints, so we make reproducibility claims at the level of labels and crossings. The legacy sweeps in Sections 4–6 (20 seeds per cell) come from the earlier platform and are not pooled with E1–E6.

4 Observed Crossings and Estimator Validation

A well-defined boundary exists at fixed training budget. At 300k steps on the supply-chain PPO substrate, collapse has a clear, tightly-estimated boundary in reward-weighting space (Figure 1): the midpoint is 0.666 $[0.631, 0.696]$. Observed proportions and Wilson intervals show the sampled support of the crossing.

"Collapse" is an operational threshold crossing, not a confirmed attractor. We tested the continuous base-stock statistic $\bar{S}$ for a distinct dynamical regime with bootstrap 95% CIs on Sarle's bimodality coefficient (Sarle, 1983) and Hartigan's dip test (Hartigan & Hartigan, 1985) in each cell of the legacy curvature-only sweep ($\lambda = 1$, α=β=0.88; $w \in \{.5, .65, .75, .85, .95\}$; $n = 20$ seeds per cell). Sarle's coefficient rises near the crossing (0.59 at w=0.65 and 0.57 at w=0.75, both above the 0.555 rule of thumb) but its CI straddles the threshold everywhere, and the dip test never rejects unimodality ($p \geq 0.30$). At $n = 20$ the dip test, calibrated against the uniform least-favorable null, has little power, so non-rejection is not evidence of unimodality; it only means the data do not establish a second regime. We therefore use *threshold-defined behavioral crossing* throughout.

5 Training-Budget Dependence

The boundary is finite-horizon. In the legacy cross-sectional sweep the linear-reward boundary slides left with budget: 0.821, 0.666, 0.622 and 0.572 at 150k, 300k, 450k and 900k requested agent transitions (Figure 1). These are separate finite-budget estimates and say nothing about within-policy recovery.

Refined single-platform cohort (E3). To follow the same policies, we trained 270 trajectories to 3.6M transitions, evaluated every 150k: linear, curvature-only and $\lambda = 7$ rewards, ten seeds, and nine mixture cells $\{.2, .4, .6, .65, .7, .75, .8, .85, 1\}$. The four cells between .6 and .85 were added because a coarser five-cell grid gave zero-width bootstrap intervals there. E3 uses the *default* optimizer configuration (single Adam step, global clip, raw rewards). Section 7 shows that in this configuration reward scale alone moves the crossing, so the E3 contrasts below include that scale pathway: they measure how transform effects evolve with budget under the default learner, not loss asymmetry. The multiplier-minus-curvature difference is positive at every primary budget but shrinks, from +0.291 [+0.233, +0.341] at 300k to +0.121 [+0.064, +0.177] at 3.6M (Table 2, Figure 7). The $\lambda = 7$ crossing falls monotonically, whereas the linear and curvature-only crossings fall until 900k and then rise again. No condition meets the pre-declared stability criterion, a 3.6M−1.8M interval within ±.05 (Table 3), so all readings are budget-conditional.

Collapse is not absorbing, and neither is non-collapse. Conditioning on the 300k label, many collapsed trajectories recover by 3.6M and some initially non-collapsed ones relapse (Table 3). The non-monotone linear and curvature crossings reflect this relapse at high w. Cell-level counts are in Appendix D.

Decoupled cohort (E5). To test the asymmetric effect over budget without the scale pathway, we repeated the horizon design with the fully decoupled learner of Section 7 (separate optimizers without clipping, normalized rewards) and arms B, A and S: 270 trajectories on the same nine cells to 3.6M. All three pre-declared predictions are met (Table 4). A−S is positive at every primary budget, from +0.234 [+0.165, +0.322] at 300k to +0.154 [+0.053, +0.256] at 3.6M, and no symmetric shift is detectable at any budget (S−B −0.056 [−0.147, +0.043] at 300k, −0.049 [−0.240, +0.099] at 3.6M). The loss-only contrast A−B is positive to 900k and not detectable afterwards. Under this learner the crossings fall to 900k and then rise as many initially coordinated trajectories relapse into collapse, and no arm stabilizes. The asymmetric effect therefore persists over the tested budgets while the absolute crossings keep moving. Figure 2 shows all

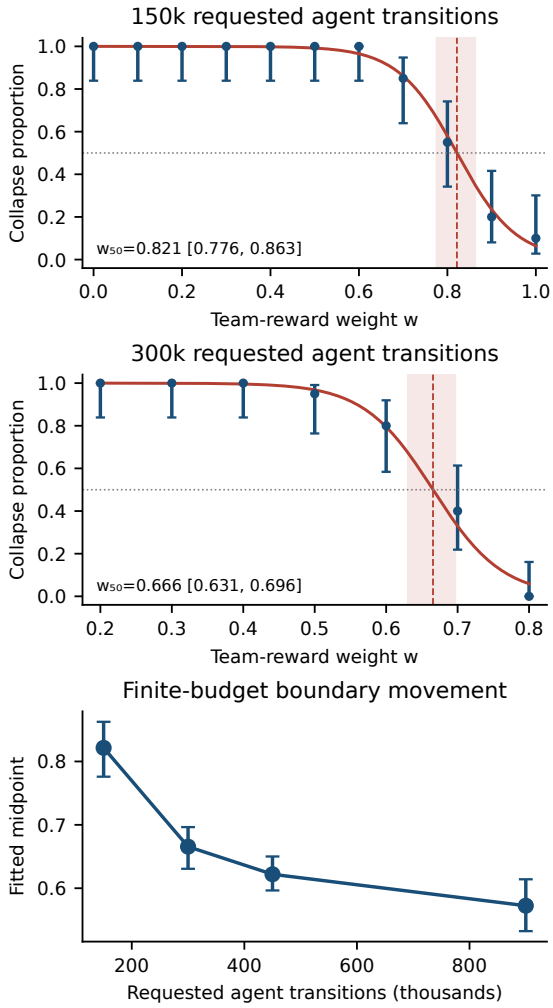


Figure 1: Finite-budget supply PPO crossings in the legacy sweep. Top and middle: observed collapse proportions with 95% Wilson intervals, Firth curves drawn only over sampled grids, midpoint lines and seed-block 95% CI bands. Bottom: independently fitted horizons; budgets count requested agent transitions.

Table 2: Refined single-platform horizon cohort (E3), default optimizer configuration (global clip, raw rewards): nine mixture cells $\{.2, .4, .6, .65, .7, .75, .8, .85, 1\}$, ten seeds, 270 trajectories trained to 3.6M and evaluated every 150k. Crossings with 95% seed-block intervals; last row is the paired contrast.

Condition	300k	900k	1.8M	3.6M
Linear	0.663 [0.628, 0.698]	0.512 [0.449, 0.584]	0.535 [0.471, 0.599]	0.578 [0.515, 0.643]
Curvature only	0.587 [0.540, 0.629]	0.473 [0.400, 0.558]	0.523 [0.453, 0.597]	0.559 [0.513, 0.612]
$\lambda = 7$	0.877 [0.843, 0.909]	0.760 [0.745, 0.775]	0.710 [0.690, 0.730]	0.681 [0.634, 0.713]
$\lambda 7 -$curvature	+0.291* [+0.233, +0.341]	+0.287* [+0.193, +0.364]	+0.187* [+0.123, +0.247]	+0.121* [+0.064, +0.177]

Table 3: Late-budget change and within-policy outcomes (E3). Stability requires the 3.6M−1.8M interval to lie within ±.05 (pre-declared for v4). Recovery: collapsed at 300k, not collapsed at 3.6M; relapse: the reverse.

Condition	3.6M−1.8M	Stable	Recovered	Relapsed
Linear	+0.043 [−0.008, +0.095]	no	14/39	8/51
Curvature only	+0.036 [−0.029, +0.105]	no	4/28	4/62
$\lambda = 7$	−0.029 [−0.071, +0.001]	no	34/76	0/14

24 checkpoints. This reading of the non-primary checkpoints is exploratory and its intervals are pointwise. The A−S interval excludes zero at 23/24 checkpoints, with point estimates between +0.099 and +0.236. The S−B interval excludes zero at 3 intermediate checkpoints (1.05–1.5M), all with S *below* B, and at no checkpoint with S above B. The default-optimizer E3 contrast is drawn for comparison; its own crossings are in Figure 7 (Appendix D).

Table 4: Decoupled horizon cohort (E5): separate actor and critic optimizers without clipping and normalized rewards; arms B, A, S; the nine E3 cells; ten seeds; 270 trajectories to 3.6M. Crossings, then paired contrasts. *: interval excludes zero.

Arm or contrast	300k	900k	1.8M	3.6M
B (curvature)	0.425 [0.344, 0.507]	0.385 [0.332, 0.439]	0.639 [0.541, 0.715]	0.732 [0.638, 0.866]
A (losses ×7)	0.603 [0.564, 0.640]	0.526 [0.452, 0.597]	0.667 [0.612, 0.731]	0.837 [0.766, 0.916]
S (both ×7)	0.369 [0.267, 0.450]	0.329 [0.237, 0.391]	0.548 [0.427, 0.646]	0.683 [0.592, 0.759]
A−S	+0.234* [+0.165, +0.322]	+0.197* [+0.086, +0.328]	+0.119* [+0.034, +0.244]	+0.154* [+0.053, +0.256]
A−B	+0.178* [+0.081, +0.271]	+0.141* [+0.064, +0.224]	+0.028 [−0.044, +0.113]	+0.105 [−0.045, +0.242]
S−B	−0.056 [−0.147, +0.043]	−0.056 [−0.146, +0.016]	−0.091 [−0.262, +0.054]	−0.049 [−0.240, +0.099]

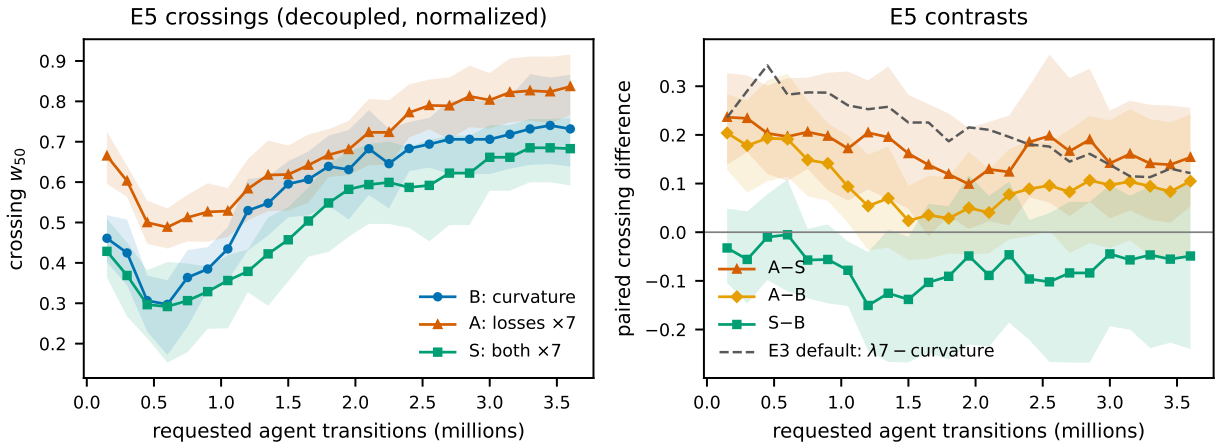


Figure 2: Decoupled horizon cohort (E5) at all 24 checkpoints. Left: crossings of B, A and S with pointwise 95% seed-block bands. Right: paired contrasts with bands; the dashed line is the default-optimizer E3 contrast $\lambda 7-$curvature (the E3 analogue of A−B, which includes the scale pathway). Nine mixture cells, ten seeds per cell and arm.

Table 5: Supply-PPO reward-transform decomposition at 300k requested agent transitions and descriptive equivalent linear-training budgets. Differences use paired seed-block draws. Equivalent budgets invert the four-point cross-sectional fit $w_{50} = 1.4824 - 0.13748 \ln(T/1000)$ and propagate the same 2,000 bootstrap replicates; they are descriptive, not causal. † denotes extrapolation below the observed 150k–900k horizon range.

Condition	Crossing and 95% CI	Versus null	Versus curvature	Equivalent budget, k
Linear null	$0.685\ [0.650, 0.715]$	–	–	$330\ [275, 406]$
Curvature ($\lambda = 1$)	$0.585\ [0.544, 0.624]$	$-0.100\ [-0.145, -0.057]$	–	$685\ [535, 942]$
$\lambda = 3$	$0.775\ [0.755, 0.790]$	$+0.090\ [+0.065, +0.115]$	$+0.190\ [+0.150, +0.229]$	$172\ [140, 211]$
$\lambda = 7$	$0.852\ [0.840, 0.865]$	$+0.167\ [+0.134, +0.203]$	$+0.268\ [+0.230, +0.306]$	$98\ [67, 128]^{\dagger}$

6 Reward-Transform Comparisons

The loss multiplier opposes curvature. In the legacy decomposition at 300k (20 seeds per cell), the linear null gives $0.685\ [0.650, 0.715]$; this is a separate run from the structural sweep of Section 4 ($0.666\ [0.631, 0.696]$), with a different w-grid, and the decomposition uses its own null. Curvature alone (α=β=0.88, the conventional prospect-theory estimate (Tversky & Kahneman, 1992), not tuned) lowers the crossing to 0.585 ($-0.100\ [-0.145, -0.057]$ vs. linear), and raising λ from 1 to 7 raises it to $0.852\ [0.840, 0.865]$ (Figure 3; per-λ values in Appendix B). Reported as a single "prospect-theoretic" treatment the two effects partially cancel. Against the curvature baseline the loss-multiplier shift is $+0.268\ [+0.230, +0.306]$. On its own, this shift cannot distinguish loss asymmetry from the larger reward magnitude the multiplier creates; Section 7 separates the two.

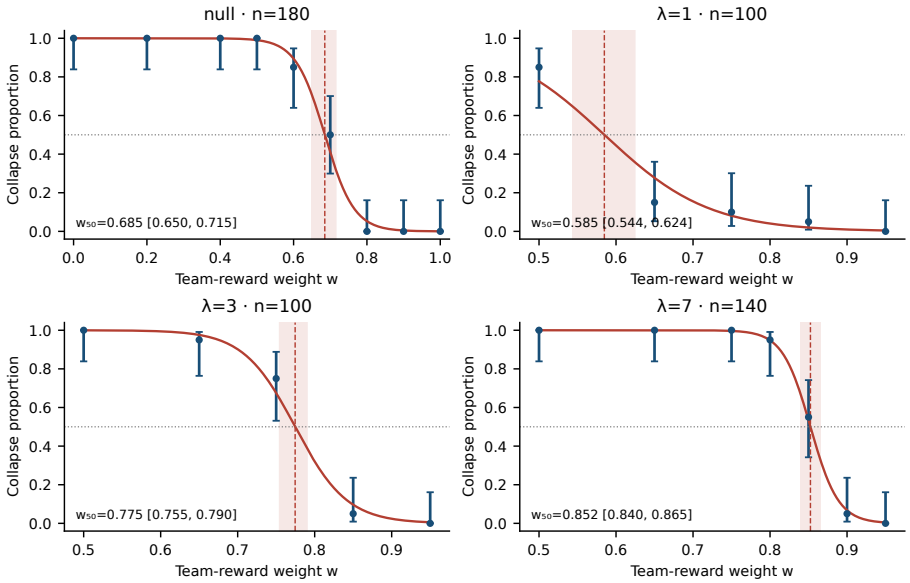


Figure 3: Legacy supply PPO null and selected transform conditions at 300k. Points show collapse counts divided by cell totals with 95% Wilson intervals; curves are Firth fits over observed grids. Vertical bands are midpoint seed-block intervals.

Equivalent linear-training budget. As a descriptive translation, inverting the legacy cross-sectional fit $w_{50} = 1.4824 - 0.13748 \ln(T/1000)$ maps curvature-only at 300k to a linear budget of 685k $[535, 942]$ and $\lambda = 7$ to 98k $[67, 128]$ (an extrapolation below the observed 150k–900k range); the repeated linear null maps to 330k $[275, 406]$ (Table 5). This mapping is not causal, but it anticipates the next section: under the default optimizer the loss multiplier behaves like a smaller training budget.

7 The Default Optimizer Masks the Asymmetric Effect

This section builds the optimizer account in five steps, summarized in Table 6: an observation, a diagnostic, an intervention that isolates one pathway, a residual effect, and an interpretation of that residual which the present experiments do not isolate. The quantities involved are distinct: the *pre-clip* gradient norm, the *applied* gradient after clipping, and the actual parameter *update* after Adam's rescaling; E6 measures all three.

Table 6: Evidence hierarchy for the optimizer account. Each level names the experiment that supports it and how far the evidence goes. "Isolated" means an intervention changes only the named quantity; "consistent" means the data agree with the explanation but an alternative is not excluded.

Level	Finding	Evidence	Status
1. Observation	Under the default configuration, symmetric $\times 7$ scaling moves the crossing (S−B $+0.357$ $[+0.249, +0.464]$).	E1, Table 8	Direct
2. Diagnostic	The critic dominates the shared clip: pre-clip critic norm 1,327 (B) vs. 11,965 (S); the actor's applied gradient for S is 0.12 of B's and its parameter update 0.39.	E1, E6, Tables 17 and 9	Direct
3. Intervention	Clipping actor and critic separately restores the actor update (0.95) and changes S−B by -0.207 $[-0.336, -0.085]$.	E6	Isolated
4. Residual	Without clipping S−B is $+0.160$ $[+0.078, +0.254]$; adding reward normalization gives -0.049 $[-0.117, +0.017]$ ($+0.050$ $[-0.035, +0.144]$ under the shared clip).	E1, E6	Direct
5. Interpretation	Candidate: the residual reflects the critic's target scale; explained variance 0.68 (B) vs. 0.30 (S) without normalization, 0.67 and 0.66 with it. Reward normalization also rescales the actor's rewards.	E1	Consistent, not isolated

Design (E1, E2). The pre-registered factorial crosses two optimizer modes with two reward modes at 300k. The optimizer modes are SB3's default global clip and separate actor and critic Adam optimizers without clipping. The reward modes are raw transformed rewards and online normalization (`VecNormalize` of rewards by the running standard deviation of discounted returns, no observation normalization). Each cell runs the four arms B, A, S and R on eight mixture cells $\{0, .2, .4, .5, .6, .7, .85, 1\}$ with seeds 42–51, for 1,280 trajectories. An addendum (E2, 320 trajectories) keeps the default clip and raw rewards but sets Adam's ϵ to 10^{-8}. The pre-declared readings were: an A−S interval excluding zero after both interventions indicates a sign-asymmetric effect; collapse of the symmetric contrasts after an intervention indicates a scale or optimizer mechanism.

Under the default configuration, symmetric scaling mimics loss asymmetry. In the default cell every minibatch exceeds the clip norm, and the critic's pre-clip gradient norm grows from 1,327 (B) to 11,965 (S) while the actor's stays near 2. The applied actor update after clipping is 0.0015 for B and 0.0002 for S (Table 17). Symmetric $\times 7$ scaling therefore moves the crossing by $+0.357$ $[+0.249, +0.464]$ relative to B, at least as far as the loss multiplier, and A−S is -0.074 $[-0.150, +0.000]$ (Table 8).

Removing the coupling removes the detectable symmetric effect but not the asymmetric one. With normalized rewards the critic's gradient falls below the clip norm, the actor carries almost all of the gradient energy, and no symmetric shift is detectable (S−B $+0.050$ $[-0.035, +0.144]$). With separate optimizers alone the symmetric $\times 7$ effect shrinks to $+0.160$ $[+0.078, +0.254]$ and the $\times 3.451$ effect is no longer detectable. With both interventions, S−B is -0.049 $[-0.117, +0.017]$, while A−B remains $+0.188$ $[+0.082, +0.266]$. The S−B intervals include zero, but they are not equivalence results: the S−B interval under both interventions extends beyond a $\pm .05$ margin. Because A and S apply the same multiplier to losses and differ only in whether gains are also scaled, A−S compares two gain-to-loss ratios (1:7 and 1:1) at

matched loss scale, and A−B isolates the loss-branch multiplier at fixed gains. A−S is positive and excludes zero under normalization (+0.237 [+0.169, +0.305]), separate optimizers (+0.174 [+0.083, +0.255]) and both together (+0.237 [+0.136, +0.346]), so the pre-declared criterion for a sign-asymmetric effect is met. How much of each reward falls on the loss branch depends on w and on the learned policy, because z_0 is fixed; near the crossings both branches are active (Table 19). The arms also differ in reward mean, not only in scale: A's transformed rewards have a negative mean at most w, whereas S and R have positive means that grow with w (Figure 8, Appendix E). A−S therefore compares gain-to-loss ratios but does not hold the first two moments fixed. The changes in each contrast relative to the default cell are in Table 16; Figure 5 summarizes them.

The artifact is not Adam's ϵ, and points to critic target magnitude. Lowering ϵ to 10^{-8} leaves the symmetric effect in place (S−B +0.258 [+0.187, +0.330]; change relative to default −0.099 [−0.243, +0.033]), contrary to our pre-declared alternative that the clipped actor gradients were being suppressed by Adam's ϵ floor. Normalizing rewards, which bounds the critic's targets, removes the detectable effect, whereas separating the optimizers only reduces it. The critic's explained variance agrees: without clipping and with raw rewards it falls from 0.68 (B) to 0.30 (S), whereas under reward normalization it is 0.67 and 0.66. Two qualifications limit this account. First, `VecNormalize` normalizes rewards, not value targets as PopArt does (van Hasselt et al., 2016). Second, the separate-optimizer arm removes clipping as well; because Adam is per-parameter, splitting one Adam into two changes nothing by itself, so this arm tests the removal of clipping, not optimizer grouping.

Clipping isolation (E6). A second pre-registered cohort (720 runs) clips actor and critic gradients separately under one Adam step, and logs, at every optimizer step, both the applied gradient and the actual parameter change (Table 9). With logging switched on, the default and no-clip modes reproduce the corresponding E1 runs exactly (480/480 labels and profits), so the logging does not alter training. Under the shared clip the actor's applied gradient for S is 0.12 of B's, but Adam partly compensates, and the actor's actual parameter update is 0.39 of B's. With per-group clipping or no clipping the ratio is 0.95 and 0.96, close to but detectably below one, so our pre-declared prediction of no difference is not met as stated. Per-group clipping reduces the symmetric effect by −0.207 [−0.336, −0.085] to +0.150 [+0.095, +0.226], and its crossings are almost identical to those without any clipping. The shared clip therefore accounts for part of the artifact, roughly 0.2 of the 0.36 shift. The remainder is unaffected by clipping, accompanies a poorly fitted critic, and disappears under reward normalization (Section 7, E1). The asymmetric contrast A−S is +0.168 [+0.077, +0.240] under per-group clipping. Figure 4 puts the three steps side by side.

The baseline crossing itself depends on the optimizer. The curvature-only crossing B is 0.538 in the default cell and 0.31–0.45 in the other cells (Table 7). An absolute w_{50} is therefore a property of the learner's optimizer configuration as well as of the reward design; comparisons are valid only within a fixed configuration.

Table 7: Optimizer × reward-normalization factorial (E1) and Adam-ϵ addendum (E2): crossings w_{50} at 300k with 95% seed-block intervals. B: curvature only; A: losses ×7; S: both branches ×7; R: both branches ×3.451 (A's pooled RMS ratio). Ten seeds per cell, eight mixture cells, one platform.

Optimizer / reward	B	A	S	R
Global clip, raw (default)	0.538 [0.461, 0.624]	0.821 [0.775, 0.864]	0.895 [0.835, 0.957]	0.794 [0.775, 0.821]
Global clip, normalized reward	0.306 [0.201, 0.413]	0.593 [0.527, 0.655]	0.356 [0.272, 0.436]	0.367 [0.324, 0.414]
Separate optimizers, raw	0.326 [0.256, 0.392]	0.661 [0.628, 0.696]	0.487 [0.412, 0.569]	0.353 [0.270, 0.442]
Separate optimizers, normalized	0.414 [0.351, 0.506]	0.602 [0.560, 0.641]	0.365 [0.282, 0.439]	0.350 [0.252, 0.438]
Global clip, raw, Adam ϵ=10^{-8}	0.450 [0.387, 0.534]	0.757 [0.729, 0.775]	0.708 [0.649, 0.770]	0.615 [0.559, 0.683]

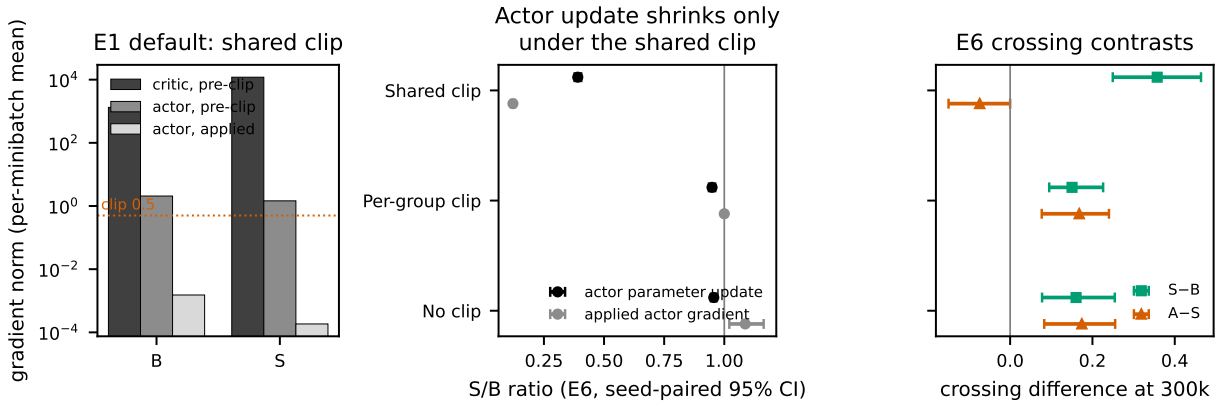


Figure 4: The shared-clip pathway. Left: in the default E1 cell (log scale) the critic's pre-clip gradient norm rises from 1,327 (B) to 11,965 (S) while the actor's stays near 2, so one global clip to 0.5 shrinks the actor's applied gradient more for S. Middle: E6 seed-paired S/B ratios of the actor's applied gradient and of its actual parameter update, with 95% intervals; a parameter-update ratio is not a gradient ratio, because Adam rescales. Right: E6 crossing contrasts. Per-group clipping restores the actor update and removes part of the symmetric shift; A−S becomes positive.

Table 8: Paired seed-block contrasts within each optimizer cell. *: 95% interval excludes zero. A−S is the pre-declared asymmetry contrast; S−B and R−B are symmetric-scaling effects.

Optimizer / reward	A−S	A−B	S−B	R−B
Global clip, raw (default)	−0.074 [−0.150, +0.000]	+0.283* [+0.181, +0.386]	+0.357* [+0.249, +0.464]	+0.256* [+0.160, +0.344]
Global clip, normalized reward	+0.237* [+0.169, +0.305]	+0.287* [+0.199, +0.386]	+0.050 [−0.035, +0.144]	+0.060 [−0.068, +0.179]
Separate optimizers, raw	+0.174* [+0.083, +0.255]	+0.334* [+0.274, +0.405]	+0.160* [+0.078, +0.254]	+0.027 [−0.024, +0.069]
Separate optimizers, normalized	+0.237* [+0.136, +0.346]	+0.188* [+0.082, +0.266]	−0.049 [−0.117, +0.017]	−0.064 [−0.176, +0.024]
Global clip, raw, Adam ϵ=10^{-8}	+0.048 [−0.013, +0.111]	+0.307* [+0.224, +0.363]	+0.258* [+0.187, +0.330]	+0.165* [+0.109, +0.219]

Table 9: Clipping isolation (E6): one Adam step with a global clip, actor and critic clipped separately, or no clipping; raw rewards; arms B, A, S on the E1 grid; ten seeds; 720 runs at 300k. Crossing contrasts, and seed-paired S/B ratios of the actor's applied (post-clip) gradient norm and of the norm of its actual parameter update. *: interval excludes zero.

Mode	S−B	A−S	Applied actor gradient S/B	Actor update S/B
Global clip (default)	+0.357* [+0.249, +0.464]	−0.074 [−0.150, +0.000]	0.121 [0.114, 0.129]	0.391 [0.378, 0.405]
Per-group clip	+0.150* [+0.095, +0.226]	+0.168* [+0.077, +0.240]	1.000 [1.000, 1.000]	0.949 [0.936, 0.962]
No clip	+0.160* [+0.078, +0.254]	+0.174* [+0.083, +0.255]	1.087 [1.021, 1.164]	0.956 [0.944, 0.967]

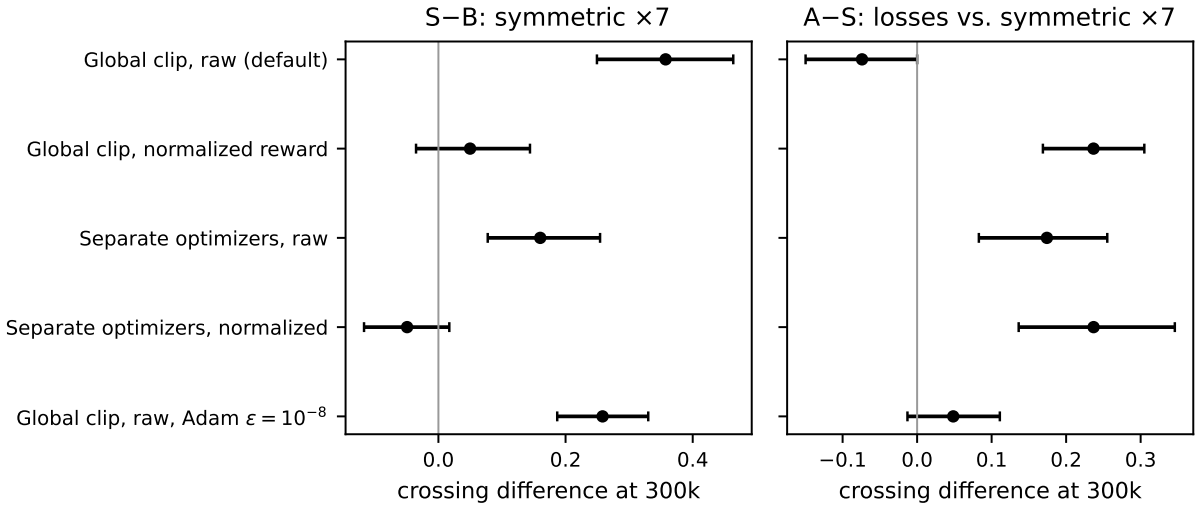


Figure 5: Symmetric-scaling (left) and asymmetric-minus-symmetric (right) crossing differences in each optimizer cell, with paired 95% seed-block intervals. Under the default configuration (top row) symmetric scaling moves the crossing as far as the loss multiplier; with normalized rewards and/or separate optimizers it does not, while A−S becomes positive.

8 A Second Environment: Level-Based Foraging

Design (E4). We repeated the arm comparison with centralized-critic MAPPO (Yu et al., 2022) on Level-Based Foraging (Christianos et al., 2020) (5×5 grid, two players, two foods), using the same SB3 update with a global clip or with separate optimizers. The local reward is the native food share minus an effort cost of 0.02 per LOAD attempt; the team reward sums native food rewards; collapse is a native team return below $\theta = 0.626$ over ten evaluation episodes, the midpoint between collapsed and coordinated team returns in an earlier calibration; the evaluation never includes the effort cost. The arms are linear, curvature only, A ($\lambda = 4$) and S (both branches ×4), on ten mixture cells with ten seeds and 2M timesteps (800 trajectories).

The asymmetric contrast replicates under both optimizers. A raises the crossing relative to S by +0.447 [+0.312, +0.559] with the global clip and +0.490 [+0.315, +0.619] with separate optimizers, and relative to curvature only by +0.543 [+0.404, +0.638] and +0.617 [+0.394, +0.767] (Table 10); the two optimizer modes do not differ detectably. The asymmetric contrast holds for collapse thresholds from 0.5 to 0.75 and under the isotonic estimator (Table 21). Here, unlike the supply chain, the critic does not dominate the clip: the actor carries 0.83–0.98 of the squared gradient norm in every arm. The loss branch is reached almost only through the LOAD cost, so the loss multiplier changes the cost of the cooperative action relative to food rewards, while S scales both and leaves their ratio unchanged; under A the share of training rewards below zero falls to a few percent at low and middle w (Table 22), because agents stop attempting loads. We therefore read the LBF result as a second environment in which an asymmetric transform moves the crossing through an incentive change, not as an independent confirmation of the supply-chain mechanism. Figure 6 shows the observed proportions behind Table 10. Several arms are far from the logistic shape, which is why Firth and isotonic crossings differ more here than on the supply chain; A's curve lies to the right of S's under both estimators.

Two pre-declared predictions are not confirmed. We had predicted that curvature would raise the LBF crossing under the global clip (P1) and that separate optimizers would shrink the curvature and symmetric shifts (P3). Under the global clip the linear arm's crossing lies below the sampled grid (Table 10), so P1 and the curvature half of P3 have positive point estimates but are not identified; the symmetric half of P3 is not met. Both predictions assumed the supply-chain clipping pathway, which LBF does not exhibit.

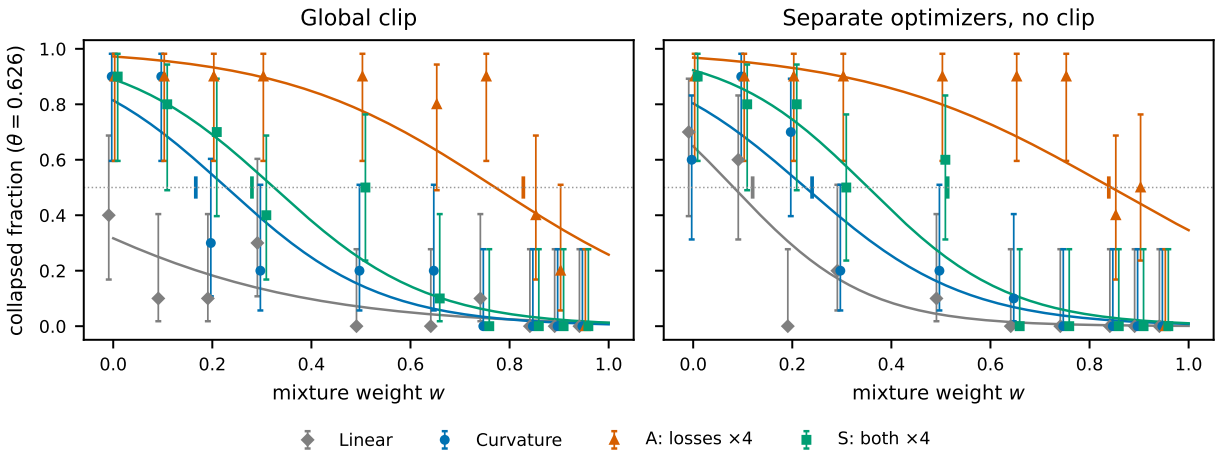


Figure 6: E4 (MAPPO on LBF): observed collapse proportions at $\theta = 0.626$ with 95% Wilson intervals (ten seeds per point; arms offset slightly in w), Firth fits (lines) and decreasing-isotonic crossings (vertical ticks at 0.5). The global-clip linear arm has no isotonic crossing in the grid and its Firth crossing lies below it (Table 10).

The stored run labels used a placeholder threshold of 0.5 instead of the declared 0.626; all E4 results here are recomputed at 0.626 from the stored team returns (deviation D4, Appendix C).

Table 10: MAPPO on Level-Based Foraging (E4) at the pre-declared label $\theta = 0.626$: crossings and paired contrasts. Ten seeds, ten mixture cells, 2M timesteps, 800 trajectories. A−S compares the gain-to-loss ratio at matched loss scale; A−curvature isolates the loss-branch multiplier. †: point estimate outside the sampled grid (the linear crossing under the global clip), so contrasts that use it are not identified. *: identified contrast whose interval excludes zero. Threshold sensitivity is in Table 21.

Optimizer	Linear	Curvature	A	S	A−S	A−curvature
Global clip	$-0.210^{\dagger}$ [−0.809, 0.031]	0.228 [0.133, 0.334]	0.771 [0.584, 0.876]	0.324 [0.179, 0.451]	$+0.447^{*}$ [+0.312, +0.559]	$+0.543^{*}$ [+0.404, +0.638]
Separate, no clip	0.082 [−0.045, 0.169]	0.226 [0.117, 0.326]	0.843 [0.623, 0.918]	0.353 [0.224, 0.468]	$+0.490^{*}$ [+0.315, +0.619]	$+0.617^{*}$ [+0.394, +0.767]

9 Discussion and Limitations

What is established. Three findings are supported at the level of evidence stated in Table 6 and Sections 5 and 8. First, a finite-budget crossing along the reward-weighting dial can be estimated with seed-paired intervals of roughly ±0.03–0.1, but it moves with training budget, does not stabilize by 3.6M transitions, and depends on the optimizer configuration. Second, under SB3's default PPO a symmetric reward multiplier moves the crossing at least as far as a loss-only multiplier, and part of this is the shared gradient clip, which E6 isolates. Third, with separate optimizers and normalized rewards, a loss-only multiplier moves the crossing relative to a symmetric control on the supply chain, through 3.6M transitions, and an asymmetric transform moves it on Level-Based Foraging by a different route. None of this is evidence about loss aversion as a psychological mechanism: the effect is a property of the transformed training signal, and in LBF it is plainly an incentive change on a costly cooperative action.

Implications for reward-transform experiments. The practical lesson is about design, not about loss asymmetry. A reward-transform comparison in actor–critic MARL should include a symmetric-scaling control at the same multiplier and should report whether actor and critic share a gradient clip and whether

rewards or value targets are normalized; without these, a change in reward scale can pass for a change in reward shape.

Implications for actor–critic optimization. The confound needs two ingredients that are SB3 PPO's defaults: an unnormalized critic and one gradient clip over actor and critic. Advantages are normalized per minibatch, so the actor's loss is nearly scale-free, but the critic's gradient grows with the reward and takes the clip budget. Cooperative PPO implementations that normalize value targets by default (Yu et al., 2022) should be less exposed; we did not test them.

Interpreting finite-budget crossings. w_{50} is a 50% point of a fitted response curve at a stated budget, learner, reward unit and label. It is not a critical point, a safe operating point or an asymptotic property. Trajectories both recover from and relapse into collapse (Table 3), so the crossing summarizes a population of training outcomes, not an attractor. Contrasts between arms within one configuration are more stable than absolute crossings: in E5 the absolute crossings fall and rise again, while A−S stays positive. A deployment requirement such as at most 5% collapse would need its own probability target and uncertainty assessment.

Limitations that bear on the main claims. *Mechanism.* The shared-clip part is isolated; the remainder is only consistent with a critic-target-scale pathway, because reward normalization also rescales the actor's rewards and we have not run a critic-only intervention such as PopArt (van Hasselt et al., 2016). *Reward moments.* A and S differ in the training reward's mean and variance as well as in the gain-to-loss ratio (Figure 8), so A−S does not hold the first two moments fixed. *Statistics.* Ten training seeds per cell give contrast intervals of about ±0.1 at 300k and up to ±0.17 at 3.6M; intervals that include zero are reported as no detectable effect, never as equivalence, and none would support equivalence at a ±0.05 margin. Intervals are pointwise; the 24-checkpoint E5 reading is exploratory. No principal contrast changes sign when any one seed is left out (Table 12).

Limitations that bear on generality. *Environments and learners.* Two small environments and one PPO implementation (SB3 2.8.0); other implementations may couple actor and critic differently. In LBF the linear arm's crossing under the global clip lies below the sampled grid, so contrasts with it are not identified, and Firth and isotonic crossings differ by up to 0.161 there, although A−S keeps its sign under both. *Budget.* No condition has a stable crossing by 3.6M, so all readings are finite-budget. *Action representation.* An earlier ablation on the supply chain found no interior crossing for PPO, QMIX or IQL on a coarse 11-level discretization of the base-stock action. *Label and grid.* Crossings are conditional on the behavioral label and the sampled grid; non-bracketed estimates are reported, not clipped.

Reproducibility. All E1–E6 runs share one platform; trained policies are not bit-reproducible across hardware and software builds, but labels agree in 221/240 (92.1%) of matched runs, so we claim reproducibility of labels and crossings, not of policies. The artifact rebuilds every reported table, figure and number from per-run records with one command (Appendix H).

Future experiments. Two experiments would close the identification gaps above: a critic-only target normalization with the actor's rewards unchanged, and a symmetric control matched to the loss-multiplier arm's reward mean and variance. Additional seeds near the decoupled crossings would narrow the null intervals. A second PPO implementation would test whether the confound is specific to SB3's update.

10 Conclusion

Reward centralization is usually treated as a direction: more team signal helps. It also has a measurable finite-budget location, w_{50}, that moves with reward transforms, training budget and the optimizer. Measured with default SB3 PPO, a symmetric reward multiplier and a loss-only multiplier cannot be told apart, because a shared gradient clip lets reward scale reach the actor's updates. With that coupling removed and rewards normalized, the loss-only multiplier moves the crossing and the symmetric one produces no detectable shift, in the tested configurations of two environments. Reward-transform comparisons in actor–critic MARL

should therefore include symmetric-scaling controls and report the optimizer coupling, gradient clipping, reward or value normalization and training budget under which they were run.

Ethical Statement

This work is entirely simulated, with no human subjects or personal data. Reported thresholds are substrate-specific and must be re-measured before use.

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A Experimental Configuration

Supply-chain environment and learner. The six-node `NetworkManagement-v1` network has 30-period episodes. Node 1 is scripted to order 20 units per period, split equally across its action entries; nodes 2–6 are controlled by one parameter-shared PPO policy whose actor sees the padded joint observation, with one vector lane per trainable node. PPO uses Stable-Baselines3 2.8.0 with `n_steps` = 128, minibatch size 256, ten optimization epochs, learning rate 3×10^{-4}, discount 0.99, GAE coefficient 0.95, policy clip 0.2, value-loss coefficient 0.5, zero entropy coefficient, normalized minibatch advantages, separate actor and critic MLPs of width 64, one Adam optimizer with SB3's implicit $\epsilon = 10^{-5}$, and global gradient max-norm 0.5. Training is CPU-only and single-threaded, with an explicit simulator random stream per environment.

Budget arithmetic. Budgets count requested agent transitions. Five lanes and 128-step rollouts add 640 transitions per rollout, and training stops at the first completed rollout at or beyond the request: 300k requested transitions execute 300,160 (469 rollouts), and each 150k checkpoint is evaluated at the first rollout boundary after it. One agent transition is one node's step, so 300,160 transitions are 60,032 joint environment periods (about 2,001 episodes).

Reward, reference point and controls. Each agent receives $x_i(w) = (1-w)r_i^{\text{local}} + wr^{\text{global}}$ before any transform. The linear condition uses $v(x) = x$; curvature-only uses $\phi(x) = \text{sign}(x)|x|^{.88}$; the loss-multiplier condition uses $v_\lambda(x) = \phi(x)$ for $x \geq 0$ and $\lambda\phi(x)$ for $x < 0$. E1 and E2 use the reference $z_0 = -2.6281398333$, fixed before any of their runs as the pooled mean of raw blended rewards from 40 untreated curvature-only calibration policies, with 7,500 samples at each of the eight E1 mixture cells (60,000 samples). With $u = x - z_0$ and $s_{\text{in}} = 1$, the arms are B= $\phi(u)$ ($s_{\text{out}} = 1$), A= $v_7(u)$ ($s_{\text{out}} = 1$), S= $7\phi(u)$ ($s_{\text{out}} = 1/7$) and R= $3.451\,\phi(u)$ ($s_{\text{out}} = 1/3.451$, where 3.451 is A's pooled transformed-reward RMS divided by B's in the earlier direct control). E3 uses reference zero.

Optimizer interventions. `separate_no_clip` builds two Adam optimizers with the default settings, one over the actor parameters (policy MLP, action head, log-standard-deviation) and one over the critic parameters (value MLP and head), applies the unchanged PPO loss, and skips gradient clipping; execution aborts if any trainable parameter is shared. `VecNormalize` divides rewards by the running standard deviation of discounted returns ($\gamma = 0.99$, $\epsilon = 10^{-8}$, reward clip 10, no observation normalization). E2 passes `optimizer_kwargs={eps: 1e-8}`. E6's `groupwise_clip` keeps one Adam optimizer and clips the actor and critic gradients to norm 0.5 separately; E6 also records, at every optimizer step and for each parameter group, the post-clip gradient norm and the norm of the parameter change, by copying the parameters before the step. Every run logs per-minibatch actor, critic and total gradient norms before and after the clipping step.

Labels. The supply-chain label is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, with $\bar{S}$ the mean over nodes 2–6 of the deterministic base-stock target $250(a_1 + 1)$ at the reset of evaluation seed $10s$, and $\bar{\pi}$ the ten-episode mean total profit on seeds $10s, \ldots, 10s+9$. The LBF label is native team return below 0.626 over ten deterministic episodes on seeds $10s, \ldots, 10s + 9$; the native return excludes the effort cost.

LBF learner. MAPPO uses SB3 PPO with a centralized-critic policy: the actor's features come from the agent's own observation and the critic's from the concatenated global state (feature width 128; actor heads $[64, 64]$, critic heads $[128, 128]$), eight parallel environment copies, `n_steps` = 256, minibatch 512, entropy coefficient 0.01, and `lbforaging` 2.0.0. In a check at 100k timesteps on one cell and seed, the global-clip runner reproduced the legacy LBF runner's team return exactly.

B Boundary Estimation and Diagnostics

For each fixed configuration we fit $P(Y = 1 \mid w) = \sigma(b_0 + b_1 w)$ by Firth logistic regression and report $w_{50} = -b_0/b_1$. A point estimate is reported only with an identified nonzero slope; observed proportions are inspected separately for adjacent cells bracketing .5. Confidence intervals use 2,000 percentile-bootstrap draws of complete training-seed blocks (multinomial weights over the ten seeds; random seed 20261015 for

E1–E6). The same weight vector is used for every condition in an experiment, so differences are seed-paired. Fits outside the grid and non-converged draws are retained and counted. Intervals are pointwise; no simultaneous statement across checkpoints or cells is made.

Logistic, probit and decreasing isotonic fits are sensitivity estimators on the same rows. The isotonic fit is weighted pool-adjacent-violators; its crossing is linear interpolation between the last cell with fitted probability at least .5 and the first below it, and a plateau at exactly .5 is reported as an interval with its midpoint used for comparison. The maximum spread among the three point estimators on the legacy supply-chain sweep is 0.018. Table 11 compares Firth and isotonic crossings for every in-grid E1–E4 fit.

Table 11: Agreement between Firth and decreasing-isotonic crossings for every in-grid fit. LBF collapse proportions are far from the logistic shape for some arms, which is why E4 shows the largest gaps; E4's A−S conclusion holds under both estimators (Table 21).

Experiment	Max |Firth−isotonic|	Median
E1/E2 (20 fits)	0.055	0.019
E3 (72 fits)	0.047	0.016
E4, $\theta = 0.626$ (7 in-grid fits)	0.161	0.044

Table 12: Leave-one-seed-out sensitivity of the principal contrasts (exploratory). Each contrast is refitted ten times on nine training seeds; the range of the ten point estimates and the number that keep the full-data sign are shown. E4 uses the pre-declared $\theta = 0.626$.

Experiment	Contrast	All seeds	Leave-one-out range	Same sign
E1 default	S−B	+0.357	[+0.325, +0.384]	10/10
E1 default	A−S	−0.074	[−0.098, −0.048]	10/10
E1 decoupled	S−B	−0.049	[−0.072, −0.031]	10/10
E1 decoupled	A−S	+0.237	[+0.200, +0.267]	10/10
E5, 3.6M	S−B	−0.049	[−0.102, −0.013]	10/10
E5, 3.6M	A−S	+0.154	[+0.134, +0.189]	10/10
E6 per-group clip	S−B	+0.150	[+0.135, +0.169]	10/10
E6 per-group clip	A−S	+0.168	[+0.148, +0.193]	10/10
E4 global clip	A−S	+0.447	[+0.423, +0.504]	10/10
E4 separate	A−S	+0.490	[+0.453, +0.533]	10/10

Table 13: Observed supply PPO collapse counts at 300k. Entries are $w : k/n$; Wilson intervals and all-setting cell records are exported in collapse_cells.csv.

Condition	Observed cells
null	0.00: 20/20, 0.20: 20/20, 0.40: 20/20, 0.50: 20/20, 0.60: 17/20, 0.70: 10/20, 0.80: 0/20, 0.90: 0/20, 1.00: 0/20
λ =1	0.50: 17/20, 0.65: 3/20, 0.75: 2/20, 0.85: 1/20, 0.95: 0/20
λ =2	0.50: 20/20, 0.65: 17/20, 0.75: 6/20, 0.85: 0/20, 0.95: 0/20
λ =3	0.50: 20/20, 0.65: 19/20, 0.75: 15/20, 0.85: 1/20, 0.95: 0/20
λ =4	0.50: 20/20, 0.65: 20/20, 0.75: 18/20, 0.85: 3/20, 0.95: 0/20
λ =5	0.50: 20/20, 0.65: 20/20, 0.75: 20/20, 0.80: 17/20, 0.85: 3/20, 0.90: 0/20, 0.95: 0/20
λ =6	0.50: 20/20, 0.65: 20/20, 0.75: 17/20, 0.80: 17/20, 0.85: 10/20, 0.90: 0/20, 0.95: 0/20
λ =7	0.50: 20/20, 0.65: 20/20, 0.75: 20/20, 0.80: 19/20, 0.85: 11/20, 0.90: 1/20, 0.95: 0/20

Table 14: Full supply PPO sweep after inclusion of all 120 completed refinement runs. Slopes are Firth estimates; no legacy Wald significance claims are retained.

λ	Runs	Midpoint and 95% CI	Slope	Usable
1	100	0.585 [0.544, 0.624]	−14.72	2000/2000
2	100	0.714 [0.683, 0.745]	−27.24	2000/2000
3	100	0.775 [0.755, 0.790]	−29.68	2000/2000
4	100	0.805 [0.780, 0.829]	−36.84	2000/2000
5	140	0.825 [0.812, 0.838]	−67.01	2000/2000
6	140	0.835 [0.816, 0.853]	−33.55	2000/2000
7	140	0.852 [0.840, 0.865]	−55.23	2000/2000

C Pre-registration and Deviations

E1–E4 were declared in a pre-registration written before any of their outcomes existed; E5 and E6 were declared later, in response to reviews of an earlier draft, again before any of their runs. Each of E5 and E6 had a pre-launch gate that reproduced a completed E1 run bit-for-bit through the new code path. Source, design and protocol hashes were frozen before E1 launched, and the E4 runner's hash before E4 launched, after a timing check. Launch scripts refused to run on any hash mismatch. The record shows four departures, none changing a design: (i) the frozen E1 runner omitted a protocol field read at evaluation, so every run crashed after training and before producing a label; the field was added with the mapping used by the earlier direct control, before any outcome existed; (ii–iii) two analysis-code errors (floating-point grid keys; the condition name "null" read as missing) raised exceptions before any fit was computed; (iv) found after the E4 analysis: the LBF label function reads its threshold from a calibration file that was absent from the frozen source and silently fell back to a placeholder of 0.5, so the stored E4 labels use $\theta = 0.5$ rather than the declared 0.626. Because every run stores its team return, all E4 results are recomputed at $\theta = 0.626$; the 0.5 analysis is kept as a sensitivity result (Table 21). The pre-declared predictions that failed are reported in the main text: the Adam-ϵ route (E2) and two of three LBF predictions (E4). An equality gate run before E1 reproduced the earlier platform's labels in 8/8 cells, but not its policies bit-for-bit, so archived runs were not reused.

D Refined Horizon Cohort

Table 15: E3 observed collapsed trajectories out of ten seeds per mixture cell at the four primary budgets. Full checkpoint records are in the archive.

Condition	Budget	w=0.2	w=0.4	w=0.6	w=0.65	w=0.7	w=0.75	w=0.8	w=0.85	w=1
Linear	300k	10	10	8	6	2	2	1	0	0
	900k	10	9	1	2	0	0	0	0	0
	1.8M	10	9	2	2	1	1	0	0	0
	3.6M	10	9	3	3	2	1	4	1	0
Curvature only	300k	10	10	6	1	0	0	0	1	0
	900k	10	7	2	1	0	0	0	0	0
	1.8M	10	8	3	3	0	0	0	0	0
	3.6M	10	10	3	3	0	0	1	0	1
$\lambda = 7$	300k	10	10	10	10	10	10	9	7	0
	900k	10	10	10	10	10	7	0	0	0
	1.8M	10	10	10	9	7	1	0	0	0
	3.6M	10	10	8	6	6	1	1	0	0

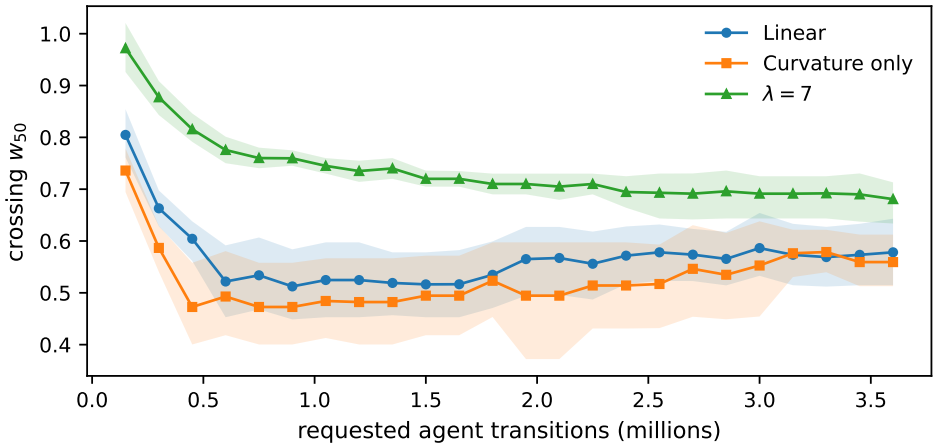


Figure 7: E3 crossings at every checkpoint (lines) with pointwise 95% seed-block bands. Default optimizer; nine mixture cells, ten seeds per cell and condition.

E Optimizer Factorial Details

Table 16: Change in each contrast relative to the default cell (global clip, raw reward), paired by seed. *: interval excludes zero.

Intervention	Δ(A−S)	Δ(A−B)	Δ(S−B)	Δ(R−B)
Global clip, normalized reward	+0.311* [+0.220, +0.402]	+0.003 [−0.130, +0.146]	−0.308* [−0.443, −0.158]	−0.195* [−0.349, −0.046]
Separate optimizers, raw	+0.248* [+0.136, +0.373]	+0.051 [−0.071, +0.161]	−0.197* [−0.366, −0.021]	−0.229* [−0.350, −0.104]
Separate optimizers, normalized	+0.311* [+0.209, +0.396]	−0.096 [−0.239, +0.034]	−0.407* [−0.565, −0.258]	−0.320* [−0.480, −0.174]
Global clip, raw, Adam ϵ=10^{-8}	+0.122* [+0.008, +0.253]	+0.023 [−0.087, +0.132]	−0.099 [−0.243, +0.033]	−0.090 [−0.201, +0.036]

Table 17: Optimizer diagnostics (means over runs of per-minibatch means). Pre-clip critic and actor gradient norms, post-clip (applied) actor norm, actor share of squared gradient norm, fraction of minibatches whose norm exceeds 0.5, and transformed-reward RMS. With separate optimizers no clipping is applied, so the applied actor norm equals the pre-clip norm.

Cell	Arm	Critic	Actor	Applied actor	Actor share	Over 0.5	Reward RMS
Global clip, raw (default)	B	1,326.81	2.07	0.0015	0.000	1.000	8.7
	A	3,425.24	1.83	0.0009	0.000	1.000	30.1
	S	11,964.76	1.46	0.0002	0.000	1.000	56.9
Global clip, normalized reward	B	0.12	4.39	0.4967	0.989	0.999	9.9
	A	0.14	4.22	0.4962	0.988	0.999	31.4
	S	0.12	4.25	0.4966	0.989	0.999	68.9
Separate optimizers, raw	B	1,285.95	4.06	4.0562	0.000	1.000	9.8
	A	3,737.64	3.63	3.6322	0.000	1.000	30.6
	S	13,923.98	4.41	4.4090	0.000	1.000	64.1
Separate optimizers, normalized	B	0.12	3.86	3.8594	0.986	0.999	9.8
	A	0.14	3.86	3.8567	0.985	0.999	31.5
	S	0.12	3.99	3.9904	0.987	0.999	69.5
Global clip, raw, Adam ϵ=10^{-8}	B	1,286.02	2.70	0.0019	0.000	1.000	9.0
	A	3,828.34	2.43	0.0011	0.000	1.000	31.1
	S	11,698.56	2.42	0.0003	0.000	1.000	59.5

Table 18: E1/E2 observed collapsed runs out of ten seeds per mixture cell at 300k.

Cell	Arm	w=0	w=0.2	w=0.4	w=0.5	w=0.6	w=0.7	w=0.85	w=1
Global clip, raw (default)	B	10	10	9	5	4	1	0	0
	A	10	10	10	10	10	10	3	0
	S	10	10	10	10	10	10	7	1
	R	10	10	10	10	10	10	1	0
Global clip, normalized reward	B	10	6	4	2	0	0	0	0
	A	10	10	9	8	4	3	0	0
	S	10	8	4	2	1	0	0	0
	R	10	10	3	1	0	0	0	0
Separate optimizers, raw	B	10	9	2	1	0	0	0	0
	A	10	10	10	10	9	2	0	0
	S	10	10	7	3	4	0	1	0
	R	10	9	3	2	0	0	0	0
Separate optimizers, normalized	B	10	10	5	2	1	0	1	0
	A	10	10	10	9	4	2	0	0
	S	10	8	6	1	0	0	0	0
	R	10	8	5	1	0	0	0	0
Global clip, raw, Adam ϵ=10^{-8}	B	10	10	5	4	1	2	0	0
	A	10	10	10	10	10	9	0	0
	S	10	10	10	9	9	4	2	0
	R	10	10	10	6	7	3	0	0

Table 19: Fraction of training rewards on the loss branch ($x - z_0 < 0$, $z_0 = -2.628$), by mixture weight, for the default and the fully decoupled E1 cells. The fraction depends on w and on the learned policy; near the crossings both branches are active.

Cell	Arm	w=0	w=0.2	w=0.4	w=0.5	w=0.6	w=0.7	w=0.85	w=1
Global clip, raw (default)	B	0.59	0.63	0.50	0.37	0.26	0.18	0.10	0.08
	A	0.59	0.65	0.58	0.57	0.55	0.49	0.13	0.07
	S	0.59	0.64	0.57	0.54	0.48	0.37	0.17	0.08
Separate optimizers, normalized	B	0.58	0.58	0.32	0.18	0.14	0.10	0.08	0.07
	A	0.58	0.62	0.51	0.38	0.24	0.17	0.09	0.08
	S	0.58	0.52	0.24	0.19	0.14	0.11	0.09	0.07

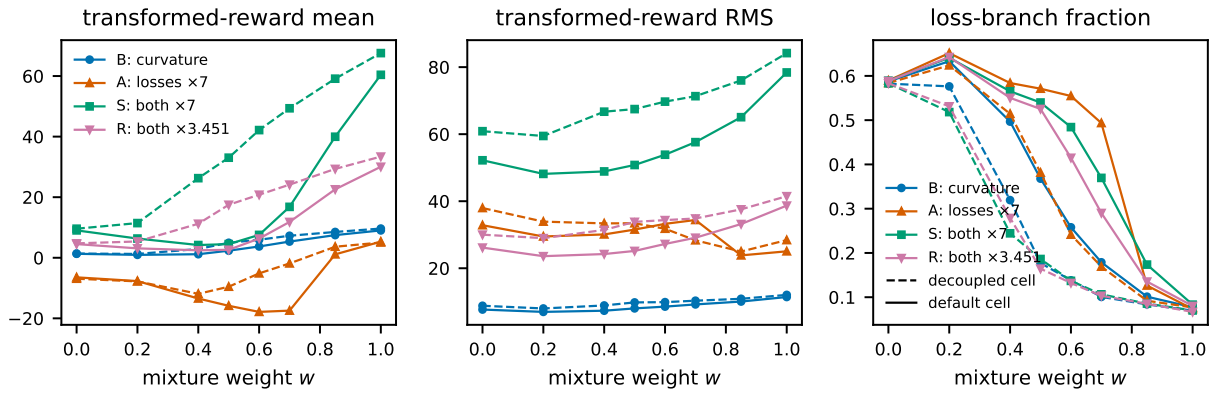


Figure 8: Transformed-reward moments over training in E1, by arm and mixture weight: mean, RMS and fraction on the loss branch (per-run means, averaged over seeds), for the default cell (solid) and the fully decoupled cell (dashed). Only per-run moments were logged, not full distributions.

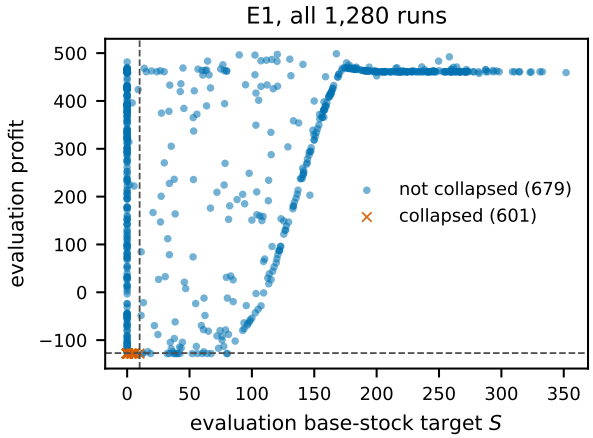


Figure 9: Collapse-label validation on all E1 runs: evaluation base-stock target $\bar{S}$ against evaluation profit, with the two label thresholds ($\bar{S} < 10$, profit ≤ -127.1) dashed. Collapsed runs sit at the lazy-policy corner.

F Level-Based Foraging Details

Table 20: E4 observed collapsed runs out of ten seeds per mixture cell (2M timesteps).

Optimizer	Arm	w=0	w=0.1	w=0.2	w=0.3	w=0.5	w=0.65	w=0.75	w=0.85	w=0.9	w=0.95
Global clip	Linear	4	1	1	3	0	0	1	0	0	0
	Curvature	9	9	3	2	2	2	0	0	0	0
	A (λ=4)	9	9	9	9	9	8	9	4	2	0
	S (×4)	9	8	7	4	5	1	0	0	0	0
Separate, no clip	Linear	7	6	0	2	1	0	0	0	0	0
	Curvature	6	9	7	2	2	1	0	0	0	0
	A (λ=4)	9	9	9	9	9	9	9	4	5	0
	S (×4)	9	8	8	5	6	0	0	0	0	0

Table 21: E4 sensitivity to the collapse threshold θ (native team return) and to the estimator. The stored run labels used $\theta = 0.5$ (deviation D4); the pre-declared $\theta = 0.626$ is primary. Last column: difference of decreasing-isotonic crossings.

θ	Optimizer	A	S	A−S (Firth)	A−S (isotonic)
0.500	Global clip	0.525 [0.188, 0.719]	0.107 [−0.065, 0.225]	+0.418 [+0.218, +0.553]	+0.575
	Separate, no clip	0.557 [0.250, 0.724]	0.077 [−0.126, 0.237]	+0.479 [+0.289, +0.613]	+0.731
0.626	Global clip	0.771 [0.584, 0.876]	0.324 [0.179, 0.451]	+0.447 [+0.312, +0.559]	+0.548
	Separate, no clip	0.843 [0.623, 0.918]	0.353 [0.224, 0.468]	+0.490 [+0.315, +0.619]	+0.325
0.750	Global clip	0.889 [0.812, 1.198]	0.519 [0.394, 0.610]	+0.370 [+0.259, +0.761]	+0.213
	Separate, no clip	0.921 [0.856, 1.097]	0.474 [0.364, 0.582]	+0.447 [+0.296, +0.694]	+0.354

Table 22: E4 fraction of training rewards below zero (reached through the LOAD effort cost), by mixture weight. Under A, agents largely stop attempting loads.

Optimizer	Arm	w=0	w=0.1	w=0.2	w=0.3	w=0.5	w=0.65	w=0.75	w=0.85	w=0.9	w=0.95
Global clip	A (λ=4)	0.012	0.014	0.013	0.014	0.018	0.020	0.021	0.042	0.059	0.112
	S (×4)	0.039	0.035	0.045	0.055	0.094	0.117	0.134	0.128	0.149	0.158
	Curvature	0.045	0.051	0.069	0.085	0.086	0.115	0.133	0.138	0.131	0.153
	Linear	0.089	0.100	0.105	0.141	0.165	0.176	0.169	0.166	0.165	0.176
Separate, no clip	A (λ=4)	0.012	0.012	0.012	0.012	0.013	0.015	0.017	0.040	0.046	0.100
	S (×4)	0.034	0.039	0.042	0.054	0.070	0.120	0.130	0.123	0.133	0.160
	Curvature	0.042	0.042	0.054	0.075	0.096	0.108	0.122	0.132	0.143	0.143
	Linear	0.066	0.076	0.118	0.137	0.161	0.172	0.169	0.162	0.157	0.175

G Computational Accounting

E1–E6 comprise 3,660 trajectories and about 1,366 single-core hours (Table 23), excluding smoke tests, equality gates and extraction. The legacy sweeps and the earlier confirmation cohorts were run separately on another platform and are not included.

Table 23: Measured v4 compute: completed Slurm array tasks, one CPU core each (single-thread PyTorch), summed elapsed time. Runs were executed at most eight concurrently.

ID	Experiment	Trajectories	Core-hours	Mean minutes per run
E1	Optimizer × normalization factorial, 300k	1,280	177.5	8.3
E2	Adam-ϵ addendum, 300k	320	44.4	8.3
E3	Refined horizon cohort to 3.6M	270	449.4	99.9
E4	MAPPO on LBF, 2M timesteps	800	143.2	10.7
E5	Decoupled horizon cohort to 3.6M	270	450.4	100.1
E6	Clipping isolation, 300k	720	101.4	8.4
	Total	3,660	1,366.3	

H Reproducibility Artifact

The anonymized archive, provided as supplementary material with this submission, contains the manuscript source; the frozen designs, pre-registration, hashes and deviation log; the executed source for E1–E6; per-run tables for every E1–E6 trajectory (protocol, label, evaluation summary and optimizer diagnostics); the extraction and analysis scripts; and the renderer that produces every v4 table, figure and number. Raw per-minibatch diagnostics and checkpoints (about 45 GB) are available on request. The legacy sweeps are accompanied by their per-run records, without a preserved execution environment.